UNIVERSITY OF COPENHAGEN
FACULTY OF HEALTH AND MEDICAL SCIENCES

Bachelor's thesis

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Cellular Senescence in Childhood Cancer Survivors. A pilot study

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TABLE OF CONTENTS

ACKNOWLEDGEMENTS **2**

ABSTRACT **3**

ABBREVIATIONS **4**

INTRODUCTION **5**

Current unmet needs 6

AIM **7**

BACKGROUND **8**

Hematopoiesis and ALL 8
ALL treatment 9
Late effects after treatment 9
Cellular senescence 12
Clinical Perspective 15
Identification of cellular senescence 15
Preliminary data from Desler Group 16

METHODS & DESIGN **17**

ALL-STAR study 17
Experimental procedure 18
PBMCs isolation and vital freezing 18
RNA purification 19
cDNA synthesis 19
RT-qPCR measurements 20
Data analysis 22

RESULTS **24**

Exclusion criteria: RNA yield and purity 24
Cq values 25
Reference gene bias 26
High Cq values 26
Exclusion criteria: Standard deviation 27
Melt curves 28
RNA expression 28

DISCUSSION **31**

Study limitations 31
Senescence detected? 32
Outlier analysis 34
Further directions 34

CONCLUSION **36**

COMPLEMENTARY DATA **37**

REFERENCES **40**

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ABSTRACT

Background: Childhood cancer survivors (CCSs) face an elevated risk of premature aging and early-onset health complications, including cardiovascular disease, endocrine dysfunction, and frailty. These late effects are often attributed to the long-term impact of chemotherapy on healthy, non-malignant cells. One proposed mechanism underlying these effects is cellular senescence – a state of cell cycle arrest accompanied by the senescence-associated secretory phenotype (SASP), which promotes chronic inflammation and tissue dysfunction. The aim of this thesis is to contribute to the investigation on whether cellular senescence is a feasible indicator of premature aging among acute lymphocytic leukemia (ALL) CCSs. This is done by assessing whether transcriptional senescence markers are detectable within the ALL-treated patients, 8-10 years post treatment.

Methods & Design: 48 PBMCs samples from 25 survivors of childhood and adolescent ALL 8-10 years post treatment, alongside 23 age- and sex-matched controls are assessed by RT-qPCR. Analyses was conducted to assess the expression of eight senescence markers IL-6, IL-8, MCP1, Lamin B1, LAP2α, NQO1, PAI1, and p21, normalized to GAPDH. Outlier analysis and non-parametric statistical testing were performed to determine group differences.

Results: RT-qPCR revealed no significant differences in gene expression between CCSs and healthy controls across markers IL-6, IL-8, MCP1, Lamin B1, LAP2α, NQO1, PAI1, and p21. Outlier exclusion revealed a statistically significant increase in IL-6 ($p = 0.021$) and a statistically decrease in IL-8 ($p = 0.033$) in CCSs compared to healthy controls.

Conclusion: The Desler Group's senescence marker panel did not detect senescence-associated gene expression in PBMCs of CCSs 8-10 years post-treatment. This may be due to sample heterogeneity, timing of sampling, or clearance of senescent cells by the immune system. While the method provides a basis for non-invasive senescence profiling, future studies should expand cohort size, refine cell-type specificity, and incorporate longitudinal sampling to better assess the clinical utility of senescence markers in CCSs.

ABBREVIATIONS

ADRCs	Adipose-Derived Regenerative Cells
ALL	Acute Lymphoblastic Leukemia
CCR2	C-C Chemokine Receptor Type 2
CCS(s)	Childhood Cancer Survivor(s)
CDK	Cyclin-Dependent Kinase
CDKi	Cyclin-Dependent Kinase Inhibitor
cDNA	Complementary DNA
Cq / Ct	Quantification Cycle / Threshold Cycle
DNA	Deoxyribonucleic Acid
DMSO	Dimethyl Sulfoxide
dsDNA	Double-Stranded DNA
HRM	High-Resolution Melt
H-Ctrl1	Healthy Control Cell Line 1
IL-6 / IL-8	Interleukin-6 / Interleukin-8
LAP2α	Lamina-Associated Polypeptide 2 Alpha
LTL	Leukocyte Telomere Length
MCP-1	Monocyte Chemoattractant Protein-1
MAPK	Mitogen-Activated Protein Kinase
NOPHO	Nordic Society of Paediatric Haematology and Oncology
NQO1	NAD(P)H Quinone Dehydrogenase 1
OS	Overall Survival
PAI1	Plasminogen Activator Inhibitor 1
PBMC(s)	Peripheral Blood Mononuclear Cell(s)
PCR / qPCR	(Quantitative) Polymerase Chain Reaction
RNA	Ribonucleic Acid
ROS	Reactive Oxygen Species
RT-qPCR	Reverse Transcription Quantitative Polymerase Chain Reaction
Rb	Retinoblastoma Protein
SASP	Senescence-Associated Secretory Phenotype
SA-β-gal	Senescence-Associated Beta-Galactosidase
scRNA-seq	Single-Cell RNA Sequencing
SD	Standard Deviation
U2OS	Human Osteosarcoma Cell Line

INTRODUCTION

Age is the number one risk factor associated with the most prevalent diseases in developed countries [1]. Research on aging focuses on the natural and irreversible decline in physiological function during adulthood [2]. Cancer is the most frequent cause of death in the Nordic countries [3], and a number of studies have documented accelerated aging among cancer survivors [4]. This premature aging subsequently leads to the onset of adverse health conditions decades earlier than their peers [5]. This has been linked to therapy-induced harm to nonmalignant cells, leaving survivors with a lifelong increased vulnerability to environmental stresses [5]. Consequently, this effect is the most pronounced in Childhood Cancer Survivors (CCSs). These late effects (health problems occurring later in life as a consequence of cancer therapy) progress throughout adulthood and have the greatest impact on survivors with the longest life expectancy post-treatment [5]. Notably, CCSs aged 35 years or older are five times more likely to develop a severe, disabling, life-threatening, or fatal health condition than their age- and sex-matched siblings [6]. The late effects encompass a wide range of morbidities, including second primary malignancies, cardiovascular conditions, chronic fatigue, respiratory conditions, metabolic syndrome, endocrinopathies, cognitive decline, musculoskeletal effects, and frailty [5-8].

Aging can be looked at in hallmarks, proposed by López-Otín et al. in 2023 as the current 12 hallmarks of aging: [2] genomic instability, telomere attrition, epigenetic alterations, loss of proteostasis, disabled macroautophagy, deregulated nutrient-sensing, mitochondrial dysfunction, cellular senescence, stem cell exhaustion, altered intercellular communication, chronic inflammation, and dysbiosis. In this thesis, the focus will be on the hallmark of cellular senescence. Cellular senescence is bidirectionally linked to the majority, if not all, of the other hallmarks of aging. This integrated theory highlights the interdependence of the hallmarks, suggesting that therapeutic targeting of a single hallmark, such as senescence, may impact several or all others [2].

The molecular mechanisms underlying premature aging in CCSs remain poorly elucidated [9]. However, cellular senescence has been proposed as a driving factor for the late effects observed in this population [4]. Today, cellular senescence is being recognized as a potential pharmaceutical target for age-related diseases, supported by the growing number of promising pre-clinical studies demonstrating decreased age-related pathologies and increased lifespan following senolytic interventions [10]. Therefore, senolytics therapies (the elimination of senescent cells) may help mitigate some of the late effects for CCSs.

Current unmet needs

The molecular mechanisms underlying premature aging in CCSs are not yet sufficiently investigated [9], making the mitigation of the late effects challenging. The development of an accessible and standardized measurement of cellular senescence in CCSs could support personalized post-treatment care by possibly enabling early identification of individuals at highest risk of the most severe late effects. This knowledge would pave the way for prospective cohort studies, aimed at tracking the trajectory of treatment-induced premature aging and its associated adverse health outcomes. Such insights could ultimately guide the personalization of preventive strategies within clinical guidelines. The first actionable step is understanding the molecular mechanisms driving premature aging in CCSs.

AIM

The aim of this bachelor's thesis is to conduct a cross-sectional, case control study investigating whether cellular senescence is a feasible indicator of premature aging among acute lymphocytic leukemia (ALL) CCSs. This study will enroll 25 survivors of childhood and adolescent ALL, 8-10 years post treatment (90% CCSs), alongside 25 age- and sex-matched healthy subjects without a history of ALL. Circulating peripheral blood mononuclear cells (PBMC)s from 50 human blood samples are analyzed to determine whether a senescence-associated gene expression profile is detectable within the ALL-treated patients, 8-10 years post treatment. To examine this, RT-qPCR will be used to measure the expression of the Desler Group's panel of 8 senescence markers, normalized to the housekeeping gene GAPDH.

BACKGROUND

Hematopoiesis and ALL

Human blood cells are derived from myeloid and lymphoid lineages, both of which originate from hematopoietic stem cells in the bone marrow. Myeloid progenitor cells differentiate into erythrocytes, leukocytes, monocytes and platelets (see Figure 1), while lymphoid progenitor cells mature into T cells, B cells, and NK cells. Leukemia is the most prevalent form of cancer affecting children [11]. Lymphoblastic leukemia arises when genetic alterations in the lymphoid lineage drive malignant transformation and subsequent uncontrolled proliferation (see Figure 1). Leukemia occurs in two different progress rates: Acute lymphoblastic leukemia (ALL) – the most common type affecting children due to its rapid onset – and chronic lymphocytic leukemia (CLL) – the most common type seen in adults [11].

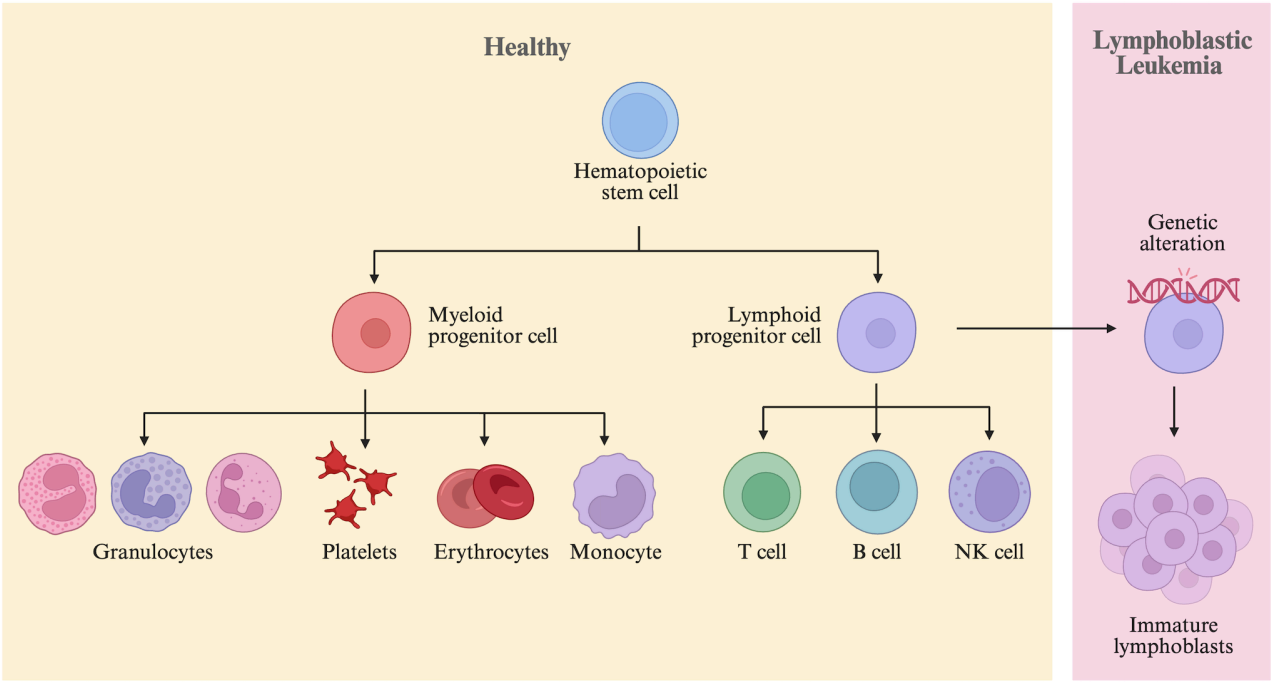


Figure 1: Healthy hematopoiesis in bone marrow and genetic alteration resulting in lymphoblastic leukemia and uncontrolled proliferation of immature lymphoblasts. Figure created with BioRender.

ALL treatment

Since 2008, ALL patients in the Nordic and Baltic countries have been treated according to the NOPHO ALL2008 protocol [12]. Currently, the 5-year overall survival (OS) rate exceeds 90% in children and 78% in adults [13] – a dramatic improvement compared to the 1950s, when ALL was a fatal disease with a median survival rate of only 4.5 months across all age groups. The treatment of ALL consists of combination chemotherapy (see Figure 2). Induction therapy is administered with steroids, while consolidation therapy includes the targeted anticancer drug asparaginase. Asparaginase specifically targets leukemic lymphoblasts, as they lack sufficient asparagine synthetase and are consequently dependent on extracellular asparagine [14]. High-risk ALL patients receive a total body irradiation of radiotherapy to prepare for a hematopoietic stem cell transplantation [12].

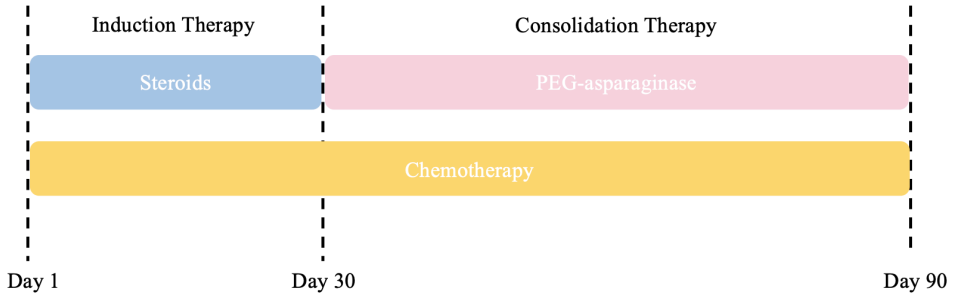


Figure 2: Overview of ALL treatment according to the NOPHO ALL2008 protocol. Day 1-30: Induction therapy, chemotherapy administered with steroids. Day 30-90: Consolidation therapy, chemotherapy in combination with PEG-asparaginase.

Late effects after treatment

The increase in lifespan among ALL patients is an oncology success story. However, healthspan remains significantly reduced compared to healthy subjects, as more than 95% of CCSs develop chronic health conditions by the age of 45 [15]. Among 24-year-old CCSs (n = 18,000) the cumulative incidence grade of health conditions is comparable to that of their 50-year-old siblings [6]. Furthermore, CCSs aged 35 years or older are 5-fold more likely than age- and sex-matched siblings to experience the onset of a new severe, disabling, life-threatening, or fatal health condition [6].

Therapy-induced health problems that occur later in life among CCSs are referred to as late effects. These include second primary malignancies [5-8, 13], cardiovascular conditions [5-8], chronic fatigue [7, 8], respiratory conditions [5-8], metabolic syndrome [5-8], endocrinopathies [5-8], cognitive decline [5, 7, 8], musculoskeletal effects [5, 7, 8]. In addition, geriatric studies have identified an increased prevalence of frailty among CCSs – characterized by strength, speed, and fatigue. In the St. Jude Lifetime Cohort Study (SJLIFE; n = 1,922) published by Ness et al. in 2018, frailty was observed at a median age of 33 years, a level comparable to that seen in individuals over 65 years of age in the general population [5] (see Figure 3).

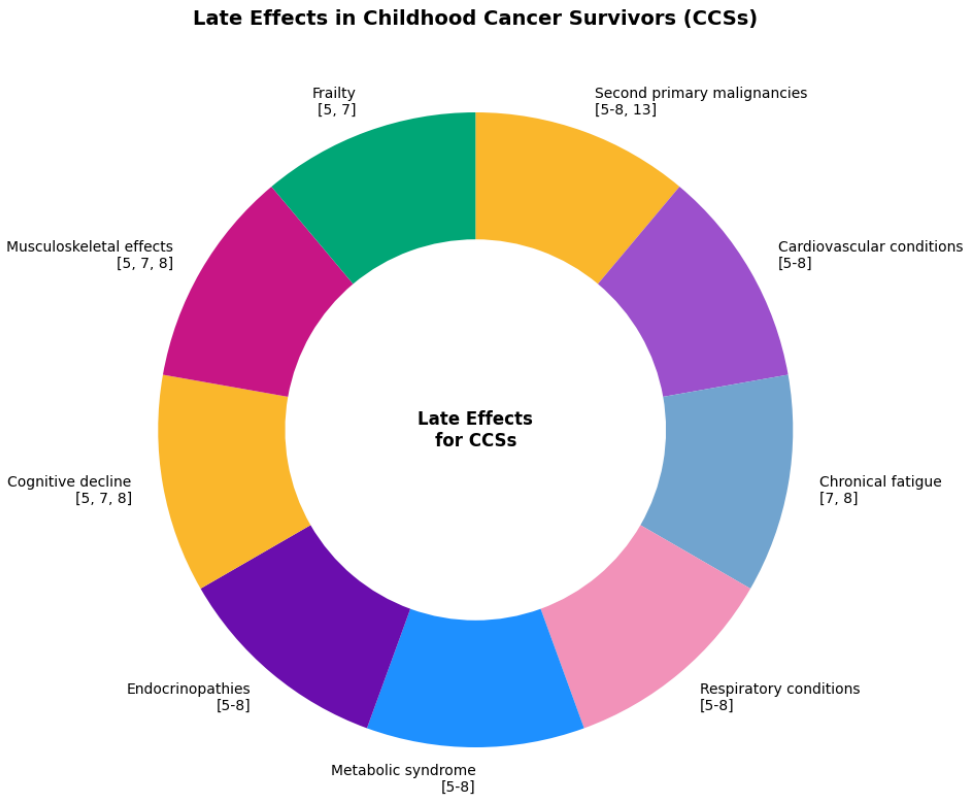


Figure 3: Overview over the most prominent late effects seen for CCSs – note this is not a percentage distribution [5-8, 13].

Cancer therapies, such as chemotherapy, suppress tumor progression primarily by inducing DNA damage that triggers cancer cell death. However, this cytotoxic effect is not limited to malignant cells as normal cells are also affected as off-targets. As a result, these treatments can induce cellular senescence and contribute to the premature aging of healthy tissues [4, 16]. The biological mechanisms underlying late effects among CCSs remain insufficiently explored. However, in 2017, Ariffin et al. investigated this phenomenon in a study of 87 asymptomatic ALL survivors with a median age of 25 years (range 18-35), all diagnosed between the ages 3 and 8 years. The recruited cohort had a gender distribution of 46% male and 54% female survivors, and an age- and sex-matched control group (n = 87) were also recruited [9]. The authors used cytometric bead array and southern blot analysis to assess plasma inflammatory cytokines and leukocyte telomere length (LTL), respectively. Analysis of covariance revealed significantly shorter LTL in survivors compared to controls ($p = 0.021$). Furthermore, comparing the survivors diagnosed before and after the age of 5, also showed significantly shorter LTL ($p = 0.043$). These findings suggest premature replicative aging in the survivor group and implicate accelerated depletion of the hematopoietic stem and progenitor cells pool. Plasma concentrations of inflammatory cytokines showed significantly elevated levels in T helper cell cytokines, which can be linked to early pathological changes. As the first generations of survivors now reach the age of 50, it is becoming increasingly evident that this population faces significant risk of premature aging [5].

It has been suggested that cellular senescence may serve as a key driver of the premature aging observed in CCSs [16]. The presence of chemotherapy-induced senescence in CCSs was investigated by Smitherman et al. in 2020 [17]. The study assessed peripheral blood T-lymphocyte $p16^{INK4a}$ (p16) expression in a cross-sectional cohort of pediatric and young adult cancer survivors (aged 18–29), compared to age-matched individuals without a history of cancer. The authors reported significantly elevated p16 expression in the survivor group, indicating an increased burden of senescent cell and supporting the hypothesis that cancer therapy induces long-lasting cellular senescence in non-malignant tissues [17]. Together, the findings of Ariffin et al. and Smitherman et al. suggest that chemotherapy accelerates aging at the cellular level. While Smitherman et al. demonstrates that chemotherapy promotes the accumulation of therapy-induced senescent cells, Ariffin et al. show that treatment leads to telomere shortening, driving replicative senescence.

Cellular senescence

Cellular senescence is characterized by permanent exit from the cell cycle and the excessive secretion of inflammatory signals – referred to as the senescence-associated secretory phenotype (SASP) [18]. The SASP encompasses a wide range of chemokines, cytokines, proteases, inhibins and activins, bioactive lipids, and microRNA [19]. Although the effects of the SASP often have inconsistent consequences for the microenvironment, in most cases the net effect is persistent chronic inflammation [2]. Senescence can be triggered by a variety of stimuli, including telomere shortening, DNA damage, oncogene activation and chemotherapy [20] (see Table 1).

Replicative senescence	The Hayflick limit reached by shortening of telomeres.
DNA damage-induced senescence	DNA damage leads to senescence or apoptosis, depending on the amplitude.
Oncogene-induced senescence (OIS)	Activation of oncogenes, e.g. RAS or BRAF, or the suppression of tumor suppressors, e.g. PTEN.
Chemotherapy-induced senescence	Anticancer drugs can induce senescence by inducing DNA damage and others by inhibition of CDKs.

Table 1: Table over four different stimuli inducing cellular senescence relevant for this thesis [20].

In senescent cells various stimuli lead to a non-proliferative state. This cell cycle withdrawal is distinct from both G_0 quiescence – since appropriate stimuli cannot restore proliferation – and terminal differentiation, as no acquisition of specific cellular functions are accompanied [21]. In senescent cells the chemotherapy, the DNA damage response (DDR), and the Arf/MDM2/p53 pathway induce the expression of the cyclin-dependent kinase 2 inhibitor (CDK2i) p21. This expression prevents the phosphorylation of the Rb family proteins, thereby inhibiting E2F transactivation and transcription of S-phase genes, ultimately resulting in cell cycle arrest [22-25] (see Figure 4). The DDR also promotes p16, a CDK4/6i, further stabilizing hypophosphorylated Rb proteins [23-25]. Additionally, in oncogene-induced senescence (OIS), reactive oxygen species (ROS) upregulation increases both NQO1 [26] and p38 MAPK expression [25]. p38 MAPK activates both the p16 and p53-p21 signaling cascades [24, 25]. NQO1 promotes p53 signaling through a ubiquitin-independent response, by protein-protein interaction guarding it from proteasomal degradation [27]. Thereby, the OIS reinforcement of the cell cycle arrest through the ROS/NQO1/p53 pathway can be seen as a fail-safe response to oncogenic stress [27] (see Figure 4).

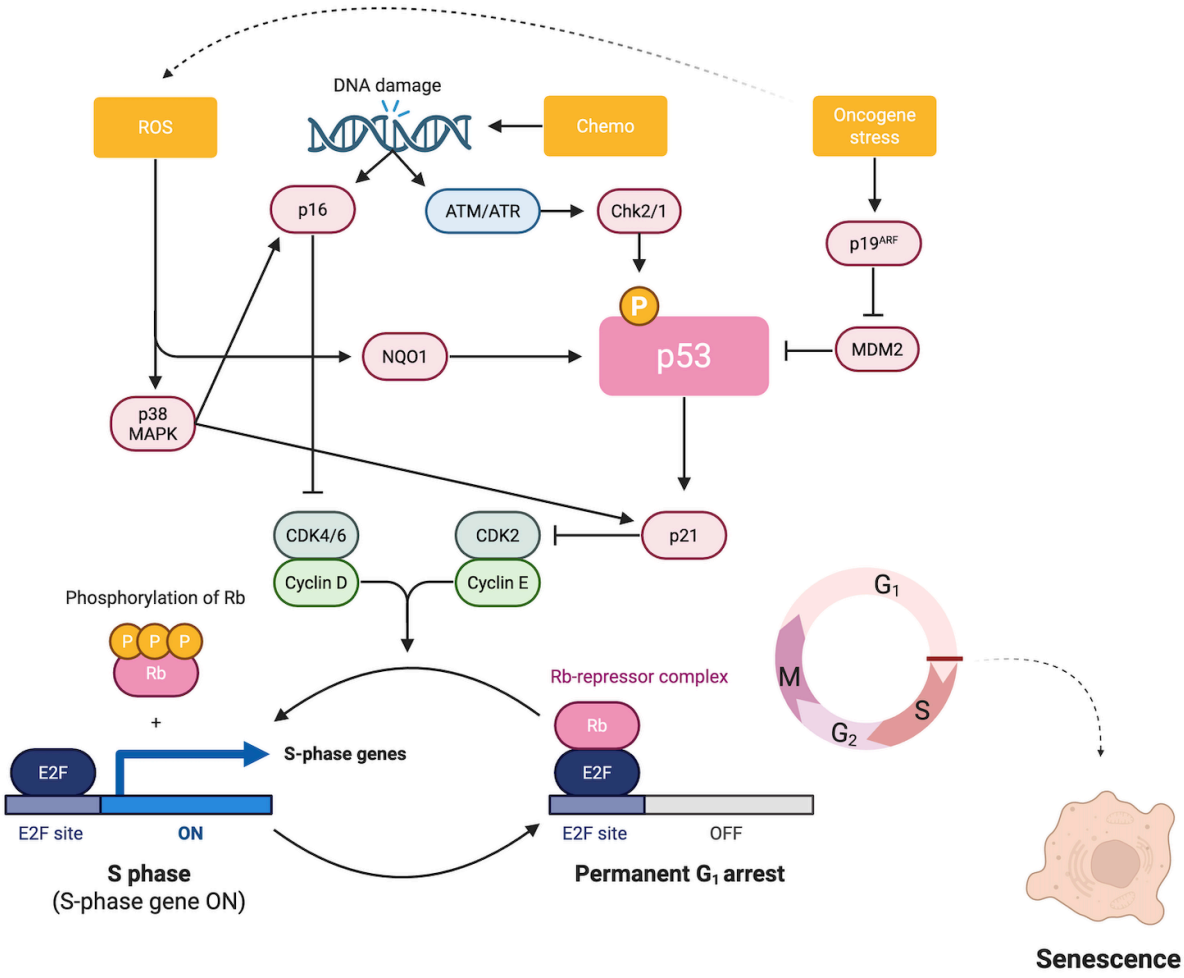
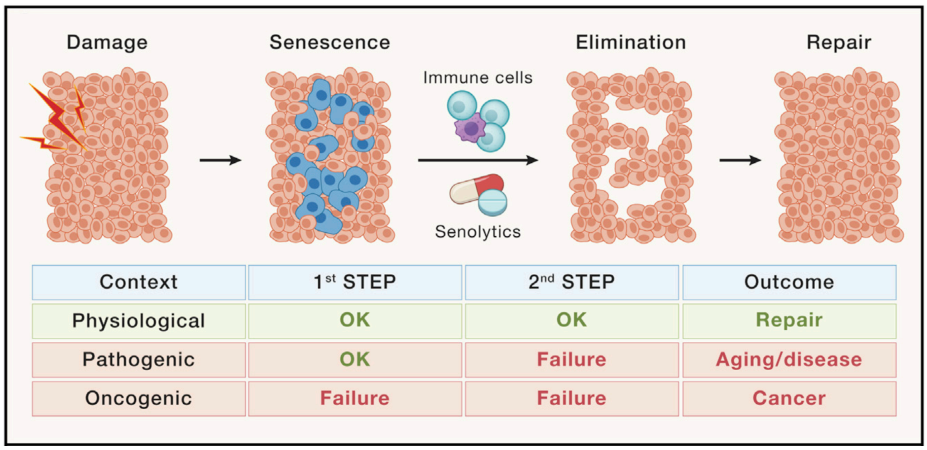


Figure 4: Overview illustrating the key regulatory networks triggered by chemotherapy, DNA damage, oxidative stress (ROS), and oncogenic stress upregulating of p16 and p21, which results in G_1 phase arrest and cellular senescence [22-27] .

The DDR induced p21 also leads to loss of H3K9 dimethylation at the interleukin-6 (IL-6) and interleukin-8 (IL-8) promoter regions, enhancing their expression [28]. IL-6 and IL-8 are among the most prominently expressed cytokines in the SASP. Additionally, monocyte chemoattractant protein-1 (MCP-1), also known as CCL2, is a key chemokine in the SASP involved in macrophage recruitment during inflammatory responses [29]. MCP-1 promotes senescence signaling via its receptor, CCR2, inducing ROS and p38-MAPK signaling. SASP factors reinforce senescence through autocrine and paracrine signaling, promoting SASP expression and DDR amplification in neighboring cells. This positive feedback loop contributes to tissue degradation and chronic microinflammation [21, 24, 25, 30, 31], and the SASP mediates the majority of the pathophysiological effects associated with senescent cells [18] [21].

The growth arrest seen in senescent cells acts as an initial protection mechanism against cell transformation and tumorigenesis [26], while the SASP recruits the immune system to eliminate the senescent cells [21] (see Figure 5). The threshold theory of the senescent cell burden hypothesizes that, when the accumulation of senescent cells exceeds the immune system's capacity for clearance, the elimination mechanism fails [19].



Context	1st STEP	2nd STEP	Outcome
Physiological	OK	OK	Repair
Pathogenic	OK	Failure	Aging/disease
Oncogenic	Failure	Failure	Cancer

Figure 5. Cellular senescence promotes tissue repair induced by environmental stress, achieved by: 1st senescence transformation and 2nd immune system elimination by SASP recruitment. If the 1st step fails, senescence induces aging and disease, and if they both fail, environmental stress induce cancer — modified from López-Otín et al., 2023 [2].

Senescence is bidirectionally linked to aging, implying that senescence both accumulates with age and contributes to the manifestation of premature aging phenotypes [18, 19, 22]. Cellular senescence is phenotypically heterogeneous but has been implicated in the development of multiple age-related diseases, including cancer [18, 19], cardiac dysfunction [2, 18, 19], respiratory diseases [2, 19, 22], cognitive decline [2, 18, 19, 22], frailty [19, 22], metabolic syndrome [2, 22], and musculoskeletal conditions [22]. As previously shown, chemotherapy has been shown to accelerate aging at the cellular level and has been implicated to induce cellular senescence [9, 32]. Among the nine late effects observed in CCSs previously described (see Figure 3), senescence is linked to seven of the nine. This overlap between the most prominent chemotherapy-induced late effects in CCSs and the senescence-induced premature aging, further supports the hypothesis that cellular senescence may be one of the primary drivers of the premature aging among CCSs.

Clinical Perspective

Senescence's bidirectional link to aging, as both a driver of tissue homeostasis decline and as a consequence of age-related processes, reveals its critical role in disease etiology. As a result, senescent cells have become a target for pharmaceutical intervention [33]. Mouse models have been utilizing to investigate the use of senolytics and have demonstrated beneficial effects on both health- and lifespan [34]. Ongoing research now focuses on targeting senescent cells to better understand their role in normal and accelerated processes [35]. The compounds are divided into senolytics – inducing senescent cell death, by targeting pathways responsible for dysregulation in senescent cells – and senomorphics, which modulate the SAPS to reduce its pathological effects.

Some of the most prominent research on the role of senolytics has been conducted at the Mayo Clinic by Baker et al., and was published in *Nature* in 2016 [34]. Using a transgenic mouse model, the researchers selectively removed p16-positive senescent cells by administering a senolytic compound from midlife onward. This intervention led to a 25-35% increase in lifespan. Furthermore, the clearance of senescent cells delayed tumor formation and mitigated age-related deterioration in multiple organs, including kidneys, heart, and adipose tissue. It also preserved exploratory behavior and physical appearance for longer. Senescence-associated β-galactosidase (SA-β-Gal) staining and electron microscopy confirmed a significant reduction in senescent cell burden across several tissues. Importantly, no adverse effects were observed on glucose metabolism, hematological parameters, or the somatotropic axis. These findings demonstrate that senescent cells contribute causally to aging phenotypes and reduces both health- and lifespan. Consequently, senolytics emerge as a promising therapeutic strategy to enhance health without inducing long-term negative effects in treated animals [34].

Identification of cellular senescence

Senescent cells and their SAPS are highly heterogeneous. Despite these diverse induction mechanisms, senescent cells may ultimately converge on a shared phenotypic profile, or, in other cases, diverge entirely [2]. Currently, the lack of one biomarker exclusively linked to senescence, or one marker expressed by all senescent cells, is a major limitation to the identification and targeting of cellular senescence both in vitro and in vivo [36]. Currently, the most used biomarker for cellular senescence is the accumulation of lysosomes, which is linked to elevated SA-β-gal activity [20].

Senescence is typically assessed in fresh tissue, ex vivo cultures and in vitro samples by SA-β-gal staining. However, the application of this assay to formalin-fixed tissue is limited, as it relies on enzymatic activity [21], making in it not ideal for clinical samples. Additionally, due to the heterogeneity of cellular senescence, SA-β-gal activity is neither required, nor exclusive to, the senescent phenotype [20].

Preliminary data from Desler Group

Unpublished data from the Desler Group suggests that assessing the expression over a panel of genes may represent the most robust approach for identifying cellular senescence both in vitro and in vivo. This gene expression profile for senescence cells is based on a literature review and confirmed in doxorubicin- and cytokine-treated U2OS, ADRCs, and H-Ctrl1 cells (see Table 2).

Biomarker	Function	Expected regulation expression in senescent cells
IL-6 & IL-8	Key cytokines in the SAPS	↑ [31, 37]
LAMINB1	Component in altered morphology	↓ [36, 38]
LAP2α	LAP2α/Lamin stabilizes pRb + LAP2α inhibits Rb/E2F-promotor	↓ [39]
MCP1	Key chemokine in the SASP	↑ [29, 40]
NQO1	ROS/NQO1/p53	↑ [26, 27]
p21	a CDKi → pRb/E2F → growth arrest	↑ [41, 42]
PAI1	↑PAI-1 sufficient to induce senescence	↑ [43, 44]

Table 2: Markers of senescence, their function and expected regulated expression in senescent cells.

Although numerous promising biomarkers besides SA-β-gal have been proposed over time in the literature, none have yet been shown to be universally expressed across all senescent cells [21]. The identification of more reliable biomarkers for monitoring senescence clearance remains crucial [20]. As a result, many – including the Desler Group – have proposed multi-marker approach [21].

METHODS & DESIGN

Design of the study is presented in Figure 6.

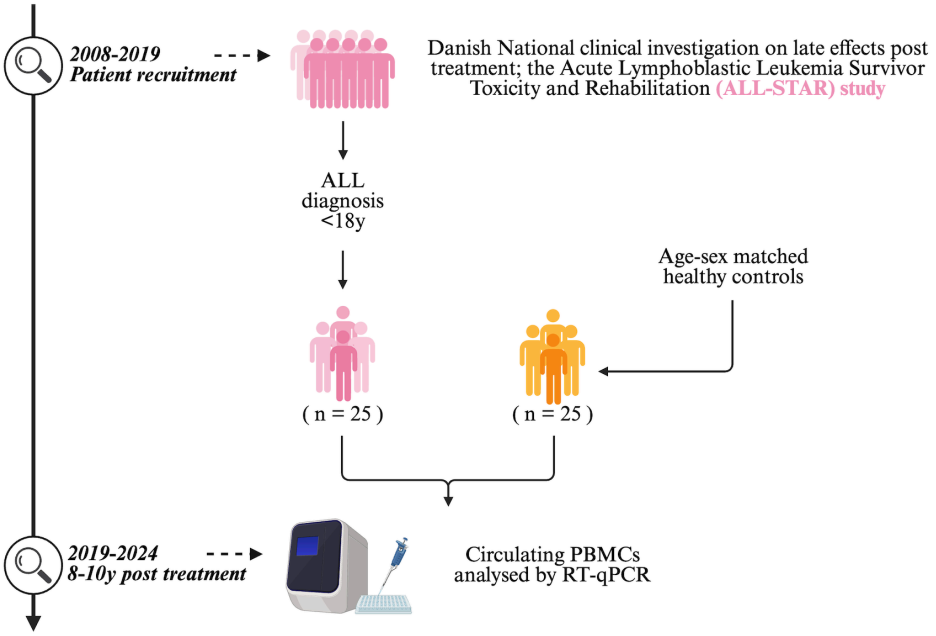


Figure 6: Study design overview and participant selection. Schematic representation over the cross-sectional case control study design enrolling CCSs of ALL 8-10 years post treatment from the ALL-STAR study + age-sex matched controls. Figure created with BioRender.

ALL-STAR study

The ALL-STAR study is a national clinical investigation focusing on late effects post treatment for ALL. All patients were treated according to the NOPHO ALL2008 protocol, ensuring uniform diagnostic procedures, treatment regimens, and risk group stratification across the cohort, and allowing for meaningful comparison of outcomes [12]. The study included patients 1-45 years of age and diagnosed with ALL between July 2008 and July 2019 [12]. The cohort design enables longitudinal follow-up of patients and represents a unique opportunity to assess disease behavior and treatment efficacy in a real-world Nordic setting with standardized therapeutic approaches. This bachelor thesis is a cross-sectional, case control study enrolling 25 survivors of childhood and adolescent ALL, 8-10 years post treatment, alongside 25 age- and sex-matched controls (see Figure 6). Since 90% of patients in the ALL-STAR study were ≤18 at diagnosis, participants in this study will be treated as CCSs [12]. The study has been approved by the Regional Ethics Committee for the Capital Region in Denmark (no. H-18035090 and H-20006359), the Danish Data Protection Agency (no. VD-2018-519).

Experimental procedure

Flowchart over the used method is presented in Figure 7.

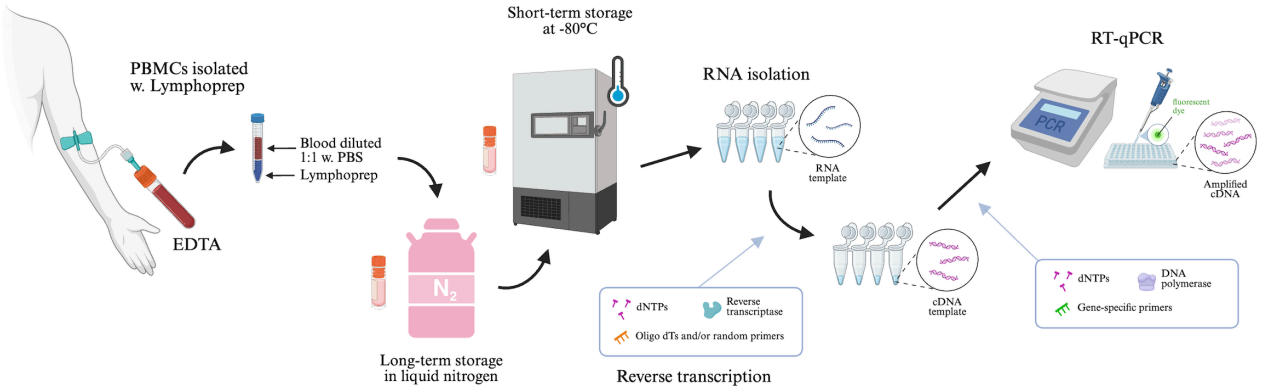


Figure 7: Flowchart over the main method used to examine the aim of this thesis. Figure created with BioRender.

PBMCs isolation and vital freezing

A total of 50 human blood samples were collected between 2019 to 2024 to be used in this study. Whole blood was collected in the ALL-STAR study in 9 mL EDTA tubes and kept on ice for up to one hour prior to processing. The whole blood was diluted 1:1 with PBS in a 50 mL tube and transferred to a Leucosep tube. 3 mL Lymphoprep was added through the center hole of the Leucosep filter, and the tubes were centrifuged at 800g for 15 min at 22°C. After centrifugation the peripheral blood mononuclear cells (PBMCs) layer was harvested immediately, to prevent cell death from prolonged Lymphoprep exposure. The PBMC layer was gently loosened from the tube wall before transferred to a 15 mL tube. The cells were washed twice in 10 mL PBS and centrifuged at 250 g for 10 min at 22°C. The cells were resuspended in buffer A (60% RPMI and 40% FBS), and 500 µL was distributed in each cryovial. The cryovials were kept on ice, while 500 µL of buffer B (80% RPMI and 20% DMSO) was added dropwise to antagonize osmotic shock. The final freezing medium contained cell culture medium, growth factors and cryoprotectant. The cryovials are placed in -80°C freezer for 24 hours before being transferred to long-term storage in liquid nitrogen.

RNA purification

The samples were transferred from Rigshospitalet and stored at -80°C. The PBMCs were thawed in a 37°C water bath and washed twice in PBS to remove the cell medium. First suspended in 10 mL PBS and centrifuged at 1000g for 5 min at 4°C, secondly resuspended in 550 µL PBS and centrifuged at 1000g for 5 min at 4°C. Total RNA was purified with the RNeasy®Plus Mini Kit (Qiagen) according to the protocol. The cells were lysed and homogenized in 600 µL RNeasy®Plus lysis buffer, inactivating RNases and ensuring isolation of intact RNA. 700 µL 70% ethanol was added to enhance RNA's binding to the membrane, and 700 µL of the sample was added to a gDNA Eliminator spin column placed in a 2 mL collection tube and centrifuged[1]. The lysate was passed through the column with a high-salt buffer to efficiently remove genomic DNA. 700 µL Buffer RW1 was added, and the sample was centrifuged[1], washing away contaminants. Buffer RW1 contains guanidine salt and ethanol to respectively remove non-specifically bound biomolecules to the silica membrane such as proteins, carbohydrates, fatty acids etc., and ensure RNA's binding to the silica membrane. The column was centrifuged twice with 500 µL Buffer RPE, centrifugation first[1] and secondly at 8500g for 2 min. Buffer RPE removes traces of salts, by earlier buffers, and ethanol to continuously make RNA remain bound. The spin column was placed in a 1,5 mL collection tube and 50 µL of RNase-free water was added to elute the purified RNA while centrifuged at 8500g for 1 min. RNA yield and purity was determined using the Epoch Microplate Spectrophotometer (BioTek).

cDNA synthesis

To perform Quantitative reverse transcription polymerase chain reaction (RT-qPCR), complementary DNA (cDNA) was synthesized from RNA template. A cDNA concentration of 25 ng/µL was chosen, therefore 500 ng RNA yield was used in a final volume of 20 µL. The volume of template RNA added was calculated by:

$$X = \frac{25\ \frac{\text{ng}}{\mu\text{L}} \cdot 20\ \mu\text{L}}{[\text{RNA}]\ \frac{\text{ng}}{\mu\text{L}}}$$

[1] Centrifugation was for 15 sek at 8500g.

The mean RNA concentration was determined by duplicate measurements on the Epoch spectrophotometer. Extracted RNA, primers and enzyme mix was thawed on ice. The calculated volume of template RNA was added to PCR tubes, in addition to 2 µL of 10× RT Reaction Premix with Oligo (dT) primer, 1.5 µL FIREScript Enzyme Mix from FIREScript®RT, and DNase/RNase-free water up to 20 µL, according to the protocol. cDNA synthesis was performed by reverse transcription reaction at 37 °C for 35 minutes, followed by enzyme inactivation at 85 °C for 5 minutes. Synthesized cDNA was diluted in ratios from 1:10 and stored at -20 °C until qPCR.

RT-qPCR measurements

qPCR allows for amplification and quantification of specific nucleic acids. The mechanism of detection is based on a fluorescence reporter molecule monitoring the DNA amplification progress. Mastermixes for all target genes combined 5× HOT FIREPol®EvaGreen®qPCR Mix Plus (ROX), forward and reverse primers, and nuclease free H_2O. The qPCR Mix contains Taq DNA polymerase, dNTPs, and DNA-binding dye. To each well in the 96-well plate 4 µL qPCR Mix, 0,5 µL of both forward and reverse primer, 11 µL nuclease-free H_2O, and 4 µL cDNA were loaded. Primer sets for target genes are listed in Table 3. Technical triplicates were made for each target gene. A negative control with 16 µL mastermix and 4 µL nuclease free H_2O were made for each primer pair/ mastermix. The plate was sealed centrifuged for 1 min at 1000g. The qPCR reaction was performed by first initial activation of the HOT FIREPol®DNA polymerase at 95◦C for 12 minutes, followed by 40 cycles of PCR. Step one was separation of the double stranded DNA (dsDNA) at 95°C for 15 seconds, breaking the helixes and leaving the single stranded DNA (ssDNA) template. Second step, 20 seconds at 60°C of annealing between primer and template DNA. In the third step, the DNA polymerase will elongate with complementary nucleotides at 20 seconds at 72°C. Melt curves analysis was performed and are included in results section to assure efficacy of primers.

Target gene	Sequence forward/ reverse	Manufacturer
GAPDH	5’-CGACCACTTTGTCAAGCTCA-3’ / 5’-GGTGGTCCAGGGGTCTTACT-3’	Eurofins Genomics
IL-6	5’- GGCACTGGCAGAAAACAACC-3’/ 5’- CACCAGGCAAGTCTCCTCATT-3’	TAG Copenhagen
IL-8	5’- TCAGAGACAGCAGAGCACAC-3’/ 5’- CTTGGCAAAACTGCACCTTCA-3’	TAG Copenhagen
LAMINB1	5’-AGCGGAAGAGGGTTGATGTG-3’ / 5’-CCTCCCATTGGTTGATCCTGT-3’	TAG Copenhagen
LAP2α	5’-ACAGTGACAATGAAGAAGGAAAG-3’ / 5’-TCCACCACCAGAGGGAGTAG-3’	TAG Copenhagen
MCP1	5’-CCCAAAGAAGCTGTGATCTTCAA-3’ / 5’-AGGGTGTCTGGGGAAAGCTA-3’	TAG Copenhagen
NQO1	5’-CTCACCGAGAGCCTAGTTCC-3’ / 5’-TCAGTGCTCTTCTGCCGAC-3’	TAG Copenhagen
$p21^{CIP\ 1}$	5’-GCCGAAGTCAGTTCCTTGTG-3’ / 5’-TTCTGACATGGCGCCTCCT-3’	TAG Copenhagen
PAI1	5’-GCCAGTGGAAGACTCCCTTC-3’ / 5’-GGGCGTGGTGAACTCAGTATAG-3’	TAG Copenhagen

Table 3: Overview of the primer pairs used for RT-qPCR.

Data analysis

BioRad software was used to export Cq values from BioRad PCR machine to evaluate relative target gene expression across Controls and CCSs. Microsoft Excel was used to calculate Cq mean for each primer target and standard deviation (SD) of the triplicate Cq measurements. An excerpt of the calculations is presented in Table 4.

	Cq #1	Cq #2	Cq #3	SD	SD <0.5?	Cq mean	ΔCq	ΔΔCq	Fold change
GAPDH	24.15073	24.14160	24.19270	0.02725	OK	24.16			
Lamin B1	31,07439	31,10023	31,06046	0,02018	OK	30.89	6.73	0.0094	1.3418

Table 4: Excerpt of Cq mean, SD, ΔCq, ΔΔCq and fold change calculations made in Excel.

Endogenous reference gene GAPDH expression was assumed constitutively expressed in the cells. Housekeeping gene normalization was performed by subtracting the Cq mean of GAPDH from the Cq mean of primer targets:

$$\Delta\mathrm{Cq} = \mathrm{Cq}_{target} - \mathrm{Cq}_{GAPDH}$$

Normalized expression was calculated as two raised to the power of minus delta Cq, reflecting the relative amount of primer target expression on a linear scale:

$$\Delta\Delta\mathrm{Cq} = 2^{-\Delta\mathrm{Cq}}$$

For relative quantification, fold changes were derived by dividing each sample's ΔΔCq by the ΔΔCq mean of the Controls group for the corresponding target gene:

$$Fold\ change = \frac{\Delta\Delta\mathrm{Cq}}{mean\ (\Delta\Delta\mathrm{Cq}_{Control})}$$

Data visualization and statistical analyses were performed using Python, utilizing open-source libraries, including numpy (*https://github.com/numpy/numpy),* pandas (*https://github.com/pandas-dev/pandas*), and matplotlib (*https://github.com/matplotlib/matplotlib*).

Due to the non-Gaussian distribution of the target genes, comparisons were conducted using a non-parametric, two-tailed Wilcoxon rank-sum test (equivalent to the Mann–Whitney U test for independent samples). This approach is robust to distributional skewness and particularly well-suited for approximately balanced and smaller group sizes. Normality was evaluated using the Shapiro–Wilk test, which revealed non-normality ($p < 0.05$) in one or both groups for the majority of genes, including IL-6, IL-8, LAP2α, MCP-1, PAI1, p21, and Lamin B1. Only NQO1 met the assumption of normality in both the Healthy subjects and CCS groups, normality test results are provided in complementary data (see Figure 14).

When an outlier analysis was conducted, z-scores was used for outlier detection defined by the criteria:

$$z = \frac{x - \mu}{\sigma}$$

x = datapoint, μ = mean of group, σ = SD of group.

Outliers were defined as $|z| > 2.0$. This approach identifies and excludes deviations from the distribution's central tendency. This definition was set to align with method used in Nature by Li et al., 2017, where outliers were identified using the same criteria [45]. Data will be presented both with and without outlier exclusion, and further outlier analysis will be conducted in results section.

RESULTS

RT-qPCR was used to test the expression level of the senescence associated targets from preliminary data by the Desler Group (see Table 2). The expression pattern of a variety of markers was tested to possible identify the Desler Group's senescence expression pattern in the ALL survivors compared to the controls. qPCR allows for a quantitative measurement of the fluorescence emitted from the reaction proportional to the amount of synthesized dsDNA. Threshold cycles (Ct) values – quantification cycle (Cq) values – are measured when the signal level shows a statistically significant increase over baseline signal. RT-qPCR is thereby a sensitive mechanism to measure gene expression on mRNA level.

Exclusion criteria: RNA yield and purity

This study is designed a cross-sectional case study, initially aiming to enroll 25 ALL CCSs, 8-10 years post treatment, alongside 25 age- and sex-matched controls (see Figure 6). Of the planned 50 patient samples, two (both from the control group) were not received. Leaving 48 samples. Additionally, samples were blindly excluded based on RNA yield and purity criteria. Sample ID correlation to group allocation (CCSs vs Control) were not known until the final month of the project. Thereby, all sample processing and analysis were performed independently of group classification. The RNA extraction and cDNA synthesis was carried out in 5 rounds. The CCSs/Control distribution was: round 1 = 29%, round 2 = 50%, round 3 = 57%, round 4 = 50%, and round 5 = 75%. Given the relatively balanced distribution across batches, this is not considered to have introduced any systematic bias to the data.

After the RNA extraction the yield and purity were determined. The absorbance ratio between 260 and 280 nm was used to assess purity. RNA yields and purities are shown in Figure 8.

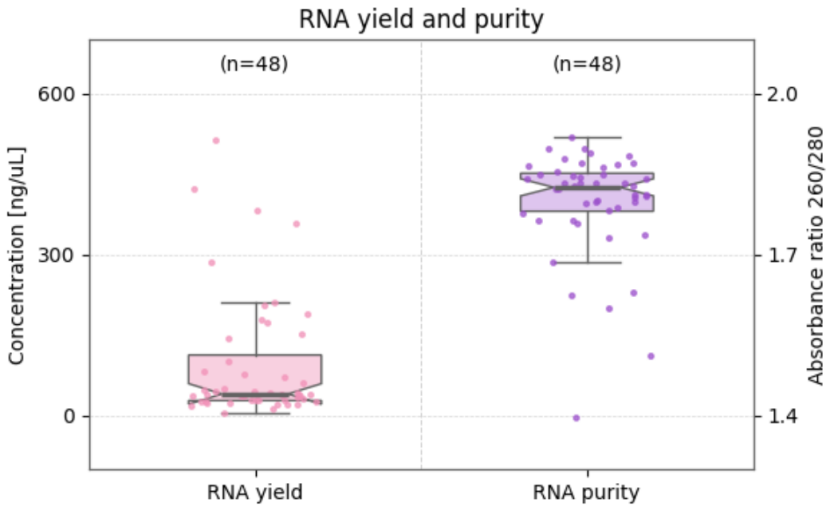


Figure 8: Notched boxplots showing RNA yield and purity of PBMC-derived samples. RNA yield (left, pink) and RNA purity (right, purple) measured in peripheral blood mononuclear cell (PBMC). RNA yield is ng/µL and RNA purity is represented by the absorbance ratio at 260/280 nm. The 260/280 ratio is used as an indicator of protein contamination in RNA preparations, with values close to 2.0 indicating high purity. Each box represents the interquartile range (IQR)(Q1 to Q3), with a mark of the median (Q2) and the notch, which reflects the 95% confidence interval of the median. Whiskers extend to the most extreme data points within 1.5×IQR from Q1 and Q3. Sample sizes (n=X) are shown above plots.

RNA extraction round 1-2 yielded a concentration with on average 170 ng/ µL RNA. The exclusion criteria were set to a 260/280 nm purity ratio below 1,8 (n = 5) or a sample concentration less than 35 ng/ µL RNA (n = 5). The overlap between the two exclusion criteria was 100%, and 5 samples were discarded from cDNA round 1-2. RNA extraction round 3-5 yielded a concentration with on average 42 ng/ µL RNA. The RNA purity on average stayed the same, and this will be elaborated on in the discussion section. Due to the significant drop in RNA yield, exclusion criteria were adjusted for cDNA rounds 3–5. Samples with RNA concentrations below 28.3 ng/µL were excluded (n = 8). Of these excluded samples, 63% overlapped with the earlier purity threshold of 1.8, leaving 35 patient blood samples for further analysis.

Cq values

The Cq values measured on the PCR machine for each gene across the two groups are presented in Figure 9. The overall similar distribution of Cq values between the two groups for the individual genes indicates properly functioning qPCR reactions. The housekeeping gene GAPDH distribution shows no statistically significant difference between groups. To ensure accurate comparison between gene expression across samples, target genes will be normalized to the expression of GAPDH.

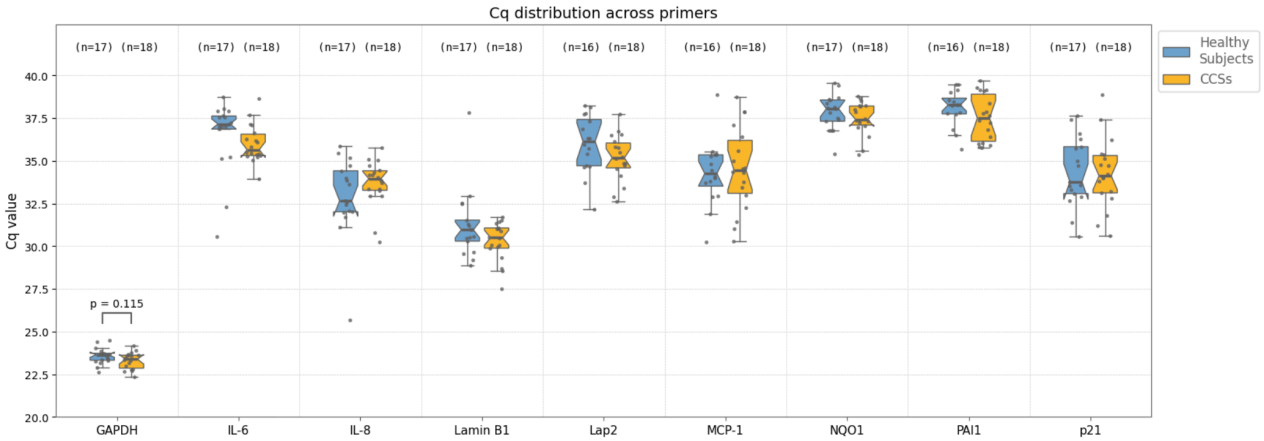


Figure 9: Notched boxplots illustrating the distribution of Cq (quantification cycle) values for nine genes measured via RT-qPCR across peripheral blood mononuclear cell (PBMC) samples from healthy controls (blue) and childhood cancer survivors (CCSs) of acute lymphoblastic leukemia (ALL) (orange). Genes measured are housekeeping gene: GAPDH, and senescence-associated gene targets: IL-6, IL-8, Lamin B1, LAP2α, MCP-1, NQO1, PAI1, and p21. Each box represents the interquartile range (IQR)(Q1 to Q3), with a mark of the median (Q2) and the notch, which reflects the 95% confidence interval of the median. Whiskers extend to the most extreme data points within 1.5×IQR from Q1 and Q3. Sample sizes (n=X) are shown above plots.

Reference gene bias

There is an indication of a lower GAPDH Cq value for CCSs (Cq mean of means = 23.56) compared to healthy subjects (Cq mean of means = 23.30) (p = 0.115). A higher GAPDH expression CCSs vs controls would eventually lead to an apparent lower target expression in CCSs after normalization. Choice of reference gene will be elaborated on in the discussion section.

High Cq values

As ssDNA turns into dsDNA during the DNA amplification EvaGreen® will undergo a conformational change allowing for DNA-binding and emitting fluorescence. A simple fluorogenic DNA-binding dye is often used amongst academic laboratories for routine PCR experiments – and in this case the dye EvaGreen® is used for forming the dye-DNA complex. EvaGreen® is binding DNA via a “release-on-demand” mechanism, consisting of two monomeric DNA-binding dyes connected by a flexible spacer. The dye is in a dynamic equilibrium between the looped inactive form and the DNA-binding capable active form, emitting fluorescence when bound to dsDNA. The looped inactive form does not bind DNA or interfere with the polymerase activity. For many commercial dyes, e.g. SYBR® Green, the optimal concentration is low limiting the PCR signal

thereby, making it unsuited for high-resolution melt curves (HRM) analysis [46, 47]. EvaGreen® was chosen for a higher threshold dye concentration resulting in more robust PCR signals and more reliable DNA melt curves. This study revealed elevated Cq values (see Figure 9). Amplification above the threshold cycle was observed in fewer than five data points. To overcome this, SYBR® Green was tested in the Desler Lab during the end of the project. Previous results were reproducible, and Cq values were on average 4 cycles lower than those obtained with EvaGreen®. However, due to the time constraint of a bachelor project SYBR® Green was not able to be implemented.

Exclusion criteria: Standard deviation

Each sample was analyzed in technical triplicates. The mean standard deviation across triplicates was 0.32 Cq, reflecting consistent pipetting technique. The distribution of SD values was: 82% with SD less than or equal to 0.5, 15% between 0.5 and 1.0, 3% between 1.0 and 1.5 (see Figure 10).

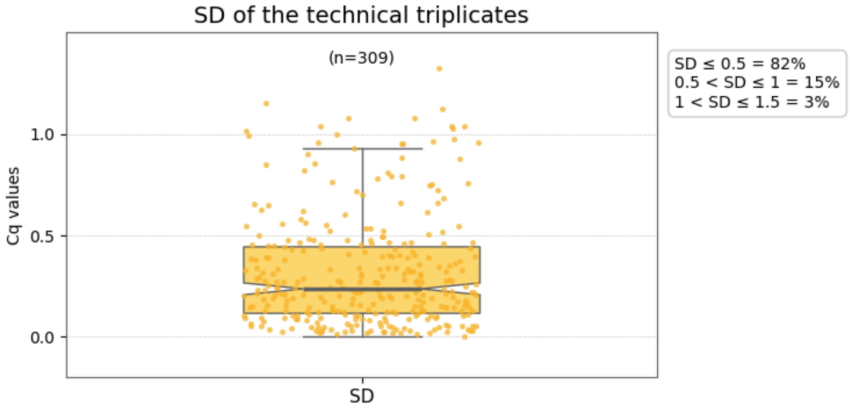


Figure 10: Standard deviation (SD) of technical triplicates across qPCR measurements. Notched boxplot displaying the distribution of SD for technical triplicates measured by RT-qPCR across all primer targets and sample groups. Each point represents the SD of a triplicate for a single gene in one sample. The distribution of SD between the groups SD ≤ 0.5, 0.5 < SD ≤ 1, and 1 < SD ≤ 1.5 are annotated. Each box represents the interquartile range (IQR)(Q1 to Q3), with a mark of the median (Q2) and the notch, which reflects the 95% confidence interval of the median. Whiskers extend to the most extreme data points within 1.5×IQR from Q1 and Q3. Sample sizes (n=X) are shown above plots.

Higher standard deviations are commonly observed for low-abundance targets [48, 49]. Given that 50% of the samples exhibited Cq values of 35 or higher (Figure 9) an elevated SD was expected. In accordance with this, an SD between 0.5 and 1.0 was considered acceptable for this pilot study. If the SD of the technical triplicates exceeded 1.0 but fell below 1.0 upon exclusion of one outlier replicate, the data point was retained (n = 7).

Melt curves

An overview over the melt curves for the 9 primers (cDNA round 2) are seen in Figure 11. Individual melt curves for each gene are seen in Figure 15 in complementary data. Smaller peaks, especially for p21 and NQO1, showed indications of primer dimer, but none exceeding threshold. No indicating off target binding were seen, but if these had been present the next step would be to include a no-RT control to test for traces of genomic DNA.

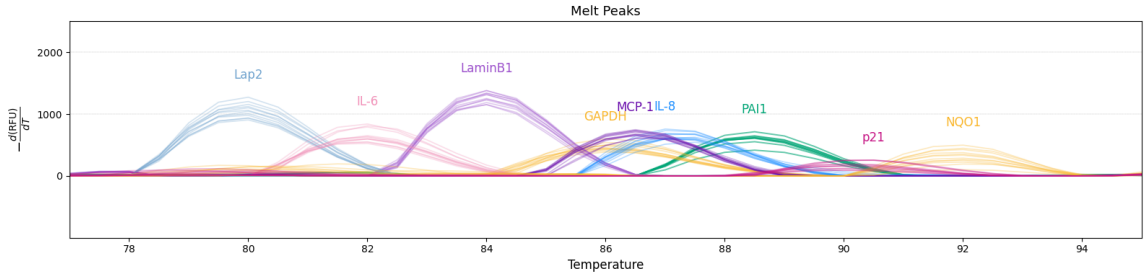


Figure 11: Melt curve profiles for senescence-associated primer targets. Overlay of melt curves generated from RT-qPCR assays for nine primer sets targeting housekeeping gene: GAPDH (dark yellow), and senescence-associated gene targets: IL-6 (light), IL-8 (dark blue), Lamin B1 (light purple), LAP2α (light blue), MCP-1 (dark purple), NQO1 (light yellow), PAI1 (green), and p21 (dark pink). Each curve represents the derivative of fluorescence change (−d(RFU)/dT) as a function of temperature.

RNA expression

RNA expression results are shown in Figure 12. The analysis reported no significance difference in comparing the healthy subjects to the ALL CCSs for the tested primers, including IL-8, IL-6, MCP-1, Lamin B1, LAP2α, NQO1, PAI1, and p21. Due to visually notable outliers seen in especially IL-6, IL-8 and MCP-1 expression within the healthy subject group an outlier analysis was performed. Outliers were defined according to the criteria outlined in the data analysis section. A total of 20 datapoint outliers were identified using these criteria, distributed as the highest 1-2 datapoint in each group (see Figure 16 in complementary data).

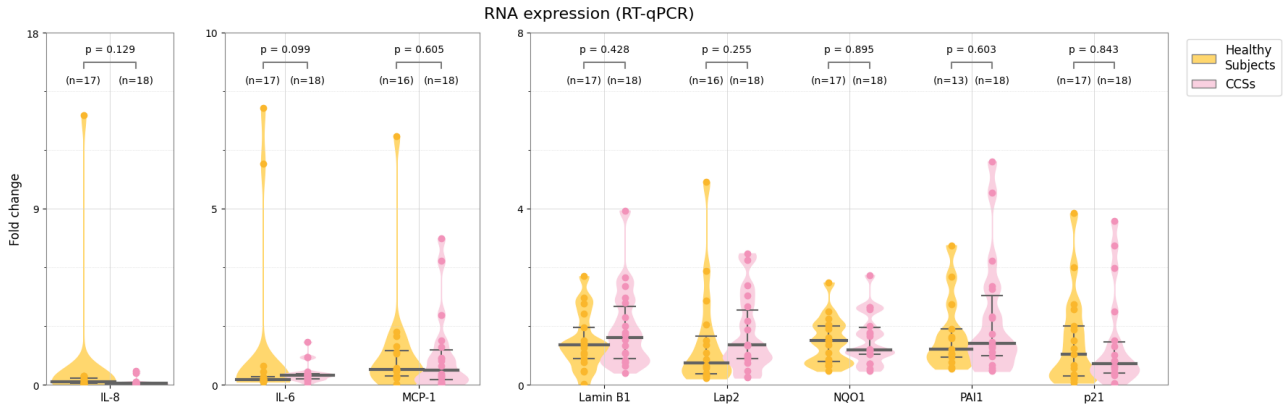


Figure 12: Gene expression levels of senescence-associated markers in PBMCs from healthy controls and childhood cancer survivors (CCSs). Violin plots showing the distribution of relative RNA expression fold change levels for eight senescence-associated genes: IL-6, IL-8, Lamin B1, LAP2α, MCP-1, NQO1, PAI1, and p21. Measured by RT-qPCR in peripheral blood mononuclear cells (PBMCs) from healthy subjects (yellow) and ALL childhood cancer survivors (CCSs, pink). Each violin shape reflects a kernel density estimation of the data distribution, scaled to a fixed width and mirrored horizontally. Densities were scaled relative to their group's maximum. Superimposed on each violin are individual data points, IQR lines (Q1 to Q3), with a mark of the median Q2. Sample sizes (n=X) and p-values from two-tailed Wilcoxon rank-sum test are shown above each plot.

With the outlier exclusion a significant increase was seen for IL-6 ($p = 0.021$) combined with a significant decrease in IL-8 ($p = 0.033$) (see Figure 13). No significant difference was seen for remaining genes, including: MCP-1, Lamin B1, Lap2, NQO1, PAI1, and p21. The impact on the p-value ranged from a minor difference for PAI1 ($\Delta p = 0.019$) to a notably larger difference for MCP-1 ($\Delta p = 0.246$). The data will be discussed in this project both with and without the outlier exclusion. The biological rationale for excluding outliers was based on the potential influence of stochastic environmental factors. While such factors in large cohort studies are more easily masked, in smaller sample sizes, even one or two outliers can disproportionately affect results. This will be further elaborated on in the discussion section.

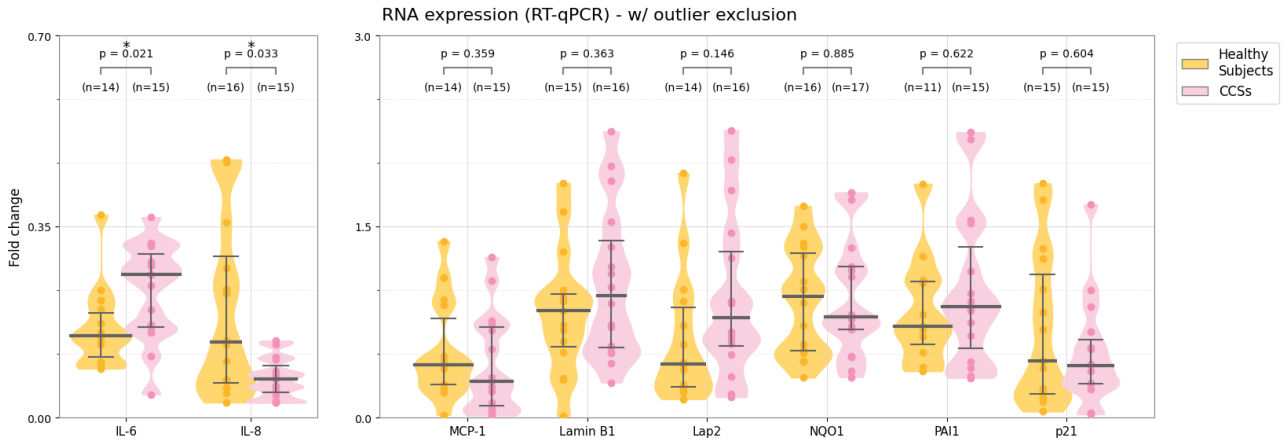


Figure 13: Gene expression levels of senescence-associated markers in PBMCs from healthy controls and childhood cancer survivors (CCSs) with outlier exclusion. Violin plots showing the distribution of relative RNA expression fold change levels for eight senescence-associated genes: IL-6, IL-8, Lamin B1, LAP2α, MCP-1, NQO1, PAI1, and p21. Measured by RT-qPCR in peripheral blood mononuclear cells (PBMCs) from healthy subjects (yellow) and ALL childhood cancer survivors (CCSs, pink). Outliers were identified using a z-score cutoff of $|z| > 2.0$. Each violin shape reflects a kernel density estimation of the data distribution, scaled to a fixed width and mirrored horizontally. Densities were scaled relative to their group's maximum. Superimposed on each violin are individual data points, IQR lines (Q1 to Q3), with a mark of the median Q2. Sample sizes (n=X) and p-values from two-tailed Wilcoxon rank-sum test are shown above each plot.

DISCUSSION

In this discussion, the obtained results will be evaluated in relation to the study aim. Preliminary studies have indicated a role of senescence as one of the drivers of the premature aging seen for CCSs. This has been proposed based on the duality of first, chemotherapy's known effect of inducing senescence and second, the notable overlap between the chemotherapy-induced late effects in CCSs and the senescence-induced premature aging. Specifically, senescence is being seen to promote adverse effects of chemotherapy and cancer relapse [32]. However, opposed to senescence many other factors have been proposed as the catalysts for the late effects experienced by CCSs [50]. The aim of this project was to be a cross-sectional, case control study to investigate whether a cellular senescence gene expression can be seen for ALL treated patients 8-10 years post treatment. First, the limitations of the study are examined. Second, the study's ability to detect cellular senescence in PBMCs from childhood ALL survivors is assessed. Finally, future directions are proposed to enhance the current findings.

Study limitations

Positive control

A notable limitation of this study is the absence of a positive control sample to validate the detection of cellular senescence markers in PBMCs. While the Desler Group's panel of senescence-associated genes has previously shown consistent expression patterns in doxorubicin- and cytokine-treated U2OS, ADRCs, and H-Ctrl1 cells, these findings have not been validated in PBMCs. Additionally, these conformations have only been confirmed in in vitro monoculture models, and not in ex vivo samples reflecting long-term in vivo chemotherapy exposure history. Incorporating a PBMC ex vivo doxorubicin-treated sample could have served as a critical assay control. This would have confirmed whether the qPCR protocol and primer panel were capable of detecting senescence-associated gene expression in the PBMC matrix specifically, benchmarking biological relevance.

Choice of reference gene

GAPDH did not show significant difference between healthy subjects and ALL CCSs, but studies regarding choice of reference gene would be relevant in the future for two reasons. First, Roy et al., 2020 studied reliable reference genes in influenza responses in PBMCs and found GAPDH to be a

poor reference gene [51]. They observed a significantly higher mean Cq value for the stimulated PBMCs vs the non-stimulated. This study only tested acute responses and therefore cannot be applied a decade post treatment. This data did not confirm a significant GAPDH expression decrease, but a possible indication of a GAPDH increase. This suggests an on average lower target gene expression for CCSs compared to their true expression. This can possibly have an undermining effect on the CCSs gene expressions that were expected to be upregulated. Given the non-significant findings of this study, the results remain unaffected. However, future research should consider reference gene validation or the use of multiple reference genes.

Senescence detected?

No statistical significance between healthy subjects and childhood ALL survivors in the cellular senescence gene expression were observed in this study. The research was performed on the basis of a senescence expression pattern earlier confirmed in U2OS, H-Ctrl1 and ADRCs cells. This study was performed on RNA extracted from PBMCs. This bulk analysis can possibly have erased subcellular expression patterns. Given the cellular heterogeneity of PBMCs, it is plausible that immune cell subsets display diverse patterns of senescence profiles, reflecting differences in their susceptibility to senescence-inducing stimuli. Accordingly, PBMCs may demonstrate varying degrees of senescence-associated transcriptional signatures. Alternatively, senescence may be more pronounced in specific subpopulations, such as T cells, while being absent or minimal in others. Consequently, this bulk approach analysis can potentially hide subset-specific expression patterns. In this study, peripheral blood was collected in EDTA tubes and PBMCs were isolated using Lymphoprep-based density gradient centrifugation. The distribution of immune cell subsets in PBMCs can vary depending on many environmental factors, and chemotherapy is seen to induce long-term compositional shifts in immune subsets [15, 52]. These effects result in lasting changes to subset proportions, e.g. chemotherapy-induced lymphopenia and suppressed B-cell maturation [52]. These limitations may influence the composition of the isolated PBMC fraction. A natural next step would be to isolate specific immune cell populations prior to RNA extraction. Flow cytometry, magnetic bead-based, and FACS-based separation would possibly allow for identification of senescent phenotypic cell groups, which cannot be resolved in bulk PBMC assays. Single-cell RNA sequencing (scRNA-seq) or bulk RNA sequencing on cell subpopulations would allow for clearer interpretation of cell-type-specific senescence signatures, determining whether therapy-induced senescence is predominantly driven by particular subpopulations.

A more narrowly defined cohort of CCSs would enhance interpretability by reducing heterogeneity. The current study design encompasses a wide array of variables, including variation in time of diagnosis, chemotherapy specification, dosing, additional therapy, and time elapsed post therapy. Expanding the cohort size and integrating a meta-analytical framework, accounting for treatment specifics, could decrease heterogeneity in results. Nonetheless, such limitations are commonly found within clinical research, given the individualized nature of cancer treatment.

Another very plausible reason for the senescence expression pattern not being detected is the timing of the sample collection. Cellular senescence is cleared by the immune system under physiological conditions. Chemotherapy is known to exert profound and complex immunosuppressive effects, both acute and long term [52]. These changes impair antigen presentation, immune surveillance, and effector function, which are all critical components for the immune system's ability to detect and eliminate senescent cells [52]. It is possible that the altered immune system immediately following chemotherapy may be more permissive for senescent cell accumulation. Sampling shortly after therapy thereby, could potentially have captured a higher load of senescence-associated transcriptional activity. In contrast, at 8-10 years post treatment, it is plausible that the immune system has partially restored. Thereby, the senescent cell burden potentially remains below the threshold capacity of the immune system for ALL CCSs, and the immune system is effective at clearing senescent cells at this stage. Conversely, sampling 20–30 years post treatment, when premature aging phenotypes and late effects such as frailty, cardiovascular disease, and cognitive impairment become more pronounced, may reveal a stronger association between cellular senescence and the late effects experienced by CCSs. In this case, the effects of chemotherapy are not confined to the immediate post-treatment window but instead initiate long-term biological reprogramming and damage responses that gradually manifest as functional decline and senescence over decades. Chemotherapy's off target effects include persistent DNA damage, mitochondrial dysfunction, and epigenetic alterations, all of which can promote senescence [52]. Possibly, senescent expression could remain undetectable the first decade post therapy. This delayed manifestation would correlate with the increase in frailty for CCSs seen at a median age of 33, as earlier mentioned [5]. Future studies should therefore aim for to longitudinally track senescence markers and correlate them with premature aging phenotypes. Such efforts would help determine whether senescence is a causative driver, or a downstream consequence of the premature aging observed in CCSs.

Outlier analysis

The outlier analysis was initially conducted due to the presence of visually notable RNA expression outliers, especially within the control group. To assess whether stochastic environmental variation within the healthy subject population could obscure a genuine expression pattern in the CCS group, an exploratory exclusion of these outliers was performed. This resulted in a significant IL-6 increase alongside a significant IL-8 decrease. The expression for the remaining genes stayed non-significant. The observed upregulation of IL-6 is well-supported in the literature as a hallmark of the SASP, with IL-6 playing a central role in reinforcing senescence and promoting inflammatory signaling cascades [37]. In contrast, the simultaneous downregulation of IL-8 stands in contradiction to prevailing models, which consistently identify both IL-6 and IL-8 as co-expressed drivers of the senescent microenvironment [31, 37] . These cytokines are known to act in a self- and cross-reinforcing manner, contributing collectively to the establishment of the pro-inflammatory, senescence-promoting environment [37]. These contradicting expressions undermines the interpretative validity of the outlier analysis. Consequently, the Desler Lab-defined senescence expression profile could not be detected in the ALL CCSs based on this analytical approach.

Further directions

SD and RNA extraction

RT-qPCR optimization was not a part of this bachelor's thesis, and primer efficiency analysis using standard curves was performed previously in the Desler Lab (data not shown). The samples received from Rigshospitalet was of very heterogeneous nature as seen by the fluctuating RNA yield. The high SD detected during this study could be induced by the low amount of cDNA template. Both Goswani et al. and Radonić et al. have found increasing variability with increasing Cq values [48, 49]. Since primer target abundance in samples were low and 50% of the samples has a Cq value of 35 or above (Figure 9), higher SDs are excepted. Therefore, upcoming experiments should focus on optimizing RNA isolation protocols to ensure consistency and higher yield, similarly to what was observed in RNA extraction rounds 1-2. Two potential reasons can be found for the significant RNA yield drop from rounds 1-2 to 3-5. Firstly, due to technical malfunctions, the QIAcube extraction employed during rounds 1–2 was not used in subsequent rounds. Instead, RNA was manually isolated in rounds 3–5, which may have reduced efficiency. Secondly, variability could have happened pre-analytical sample handling. Blood samples were collected at

Rigshospitalet over a five-year period (2019–2024), and as analysis was conducted according to chronological sample ID, it is possible that procedural changes at Rigshospitalet could have introduced a lower cell count in samples. However, as RNA purity measurements remained consistent, it is unlikely that this variability introduced systematic bias into the expression data, apart from potentially increasing SD.

Clinical Relevance

Postulating, that cellular senescence plays a role in the late effects seen for CCSs, the ability to detect senescence in PBMCs would be of significant clinical relevance. A blood sample is both non-invasive, cost-effective, and in many cases already part of routine check-up post treatment. If a correlation between senescent cell accumulation and the severity of late effects were to be established, such measurements could serve as predictive or prognostic tools in survivorship care. However, to hold clinical utility, this relationship must also be validated at a protein level, excluding post-transcriptional and translational regulation. Additionally, a quantification of circulating SASP factors in serum, would be able to measure the pathophysiological activity of the senescence cell burden. The accumulated damage caused by cancer and its therapy may be the most pronounced in CCSs. This results from its off-target effects inducing lifelong increased vulnerability to environmental stresses [5]. Thereby, CCSs are possibly the patient population that stands to benefit the most from preventive strategies against late effects. Thereby, the emerging field of senolytics may hold greatest potential of advancing health- and lifespan for CCSs. Although early clinical studies have highlighted the role of cellular senescence in disease progression, the field remains in its early stages. The prospects of introducing senolytics as a novel therapeutic approach for serious and age-related diseases – where currently no effective interventions exist – is compelling. However, the existing clinical evidence remains insufficient to support widespread application at this stage.

CONCLUSION

This study explored the feasibility of using a transcriptional senescence marker panel to detect cellular senescence in PBMCs derived from ALL CCSs. This panel have shown robust expression patterns in in vitro doxorubicin- and cytokine-induced senescence models such as U2OS, ADRCs, and H-Ctrl1 cells. Its application to clinically relevant PBMC samples did not reveal significant expression in ALL CCSs 8-10 years post-therapy compared to healthy controls. This may reflect a combination of heterogenous PBMC samples and timing of the sample collection. Nevertheless, this study established a methodological platform for RT-qPCR-based senescence profiling in PBMCs. From a clinical standpoint, a widely accessible and validated blood-based senescence assay would possibly be valuable for assessing long-term treatment effects in cancer survivors. Future studies should therefore aim to increase cohort size, decrease variabilities and incorporate longitudinal sampling. Should a correlation between senescent cell accumulation and late effect severity be confirmed, such biomarkers could function as predictive or prognostic indicators in survivorship care.

COMPLEMENTARY DATA

GENE	GROUP	SHAPIRO P-VALUE	Gaussian distribution
Lamin B1	Ctr.	0.332653	**Yes**
	CCSs	0.016676	No
Lap2	Ctr.	0.000259	No
	CCSs	0.037609	No
IL-6	Ctr.	3.83e-07	No
	CCSs	0.000283	No
IL-8	Ctr.	3.97e-08	No
	CCSs	3.6e-06	No
MCP-1	Ctr.	2.19e-06	No
	CCSs	0.000137	No
NQO1	Ctr.	0.14419	**Yes**
	CCSs	0.099673	**Yes**
p21	Ctr.	0.004856	No
	CCSs	0.000279	No
PAI1	Ctr.	0.005802	No
	CCSs	0.001492	No

Figure 14: Assessment of normality in RNA expression data using the Shapiro–Wilk test. Table summarizing the Shapiro–Wilk test results for Gaussian distribution of RNA expression data across healthy controls and childhood cancer survivors (CCSs) for eight senescence-associated genes: Lamin B1, LAP2α, IL-6, IL-8, MCP-1, NQO1, p21, and PAI1. The p-values indicate whether the distribution of gene expression values within each group conforms to a normal distribution.

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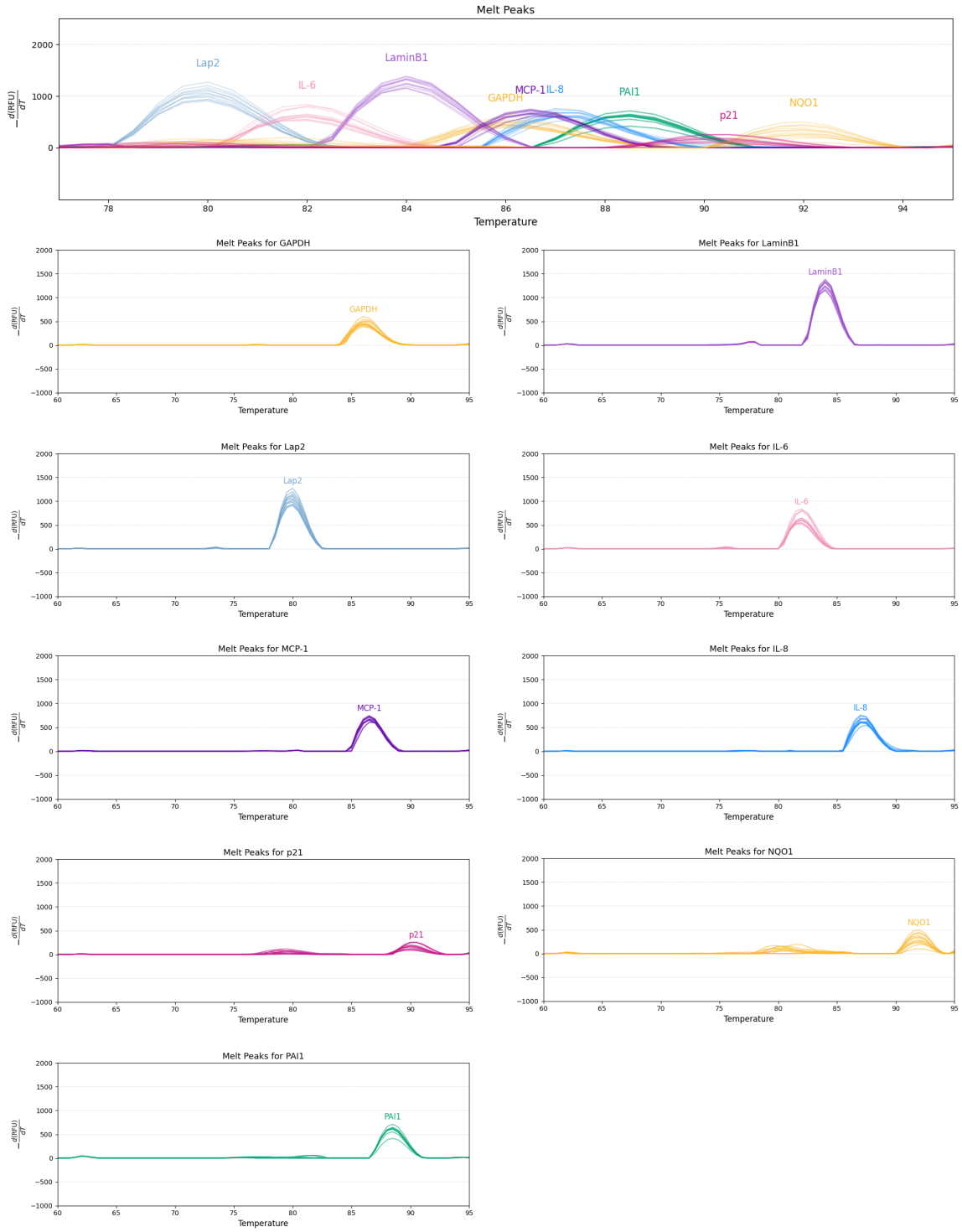


Figure 15: Melt curve analysis confirming specificity of primer pairs used for RT-qPCR. Top panel: Overlay of high-resolution melt (HRM) curves for all nine primer sets used in the study: GAPDH, IL-6, IL-8, Lamin B1, LAP2α, MCP-1, NQO1, PAI1, and p21. Melt peaks are derived from the derivative of fluorescence over temperature (−d(RFU)/dT) and plotted as a function of temperature. Bottum panel: Individual melt curve profiles for each gene.

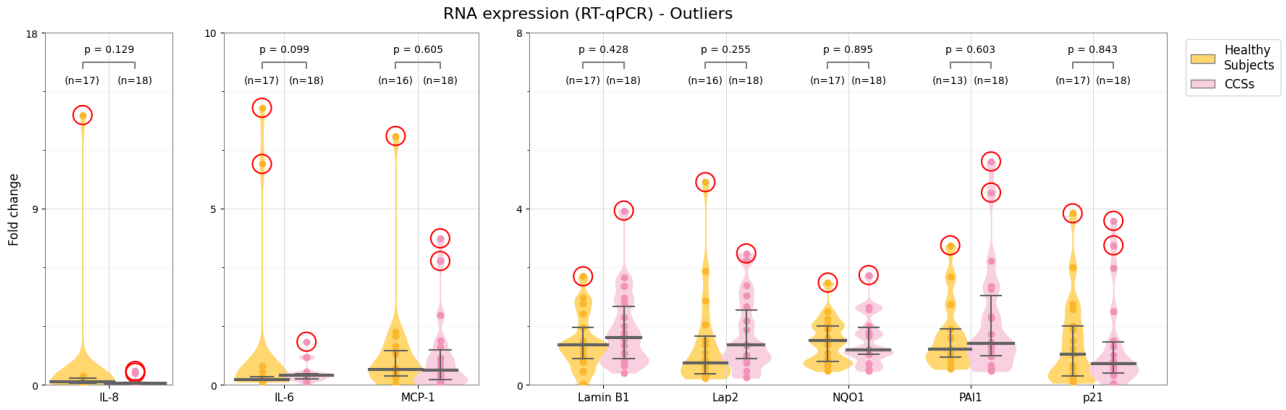


Figure 16: RNA expression of senescence-associated genes in PBMCs with outlier for each group identified. Violin plots showing relative RNA expression across healthy controls (yellow) and childhood cancer survivors (CCSs, pink) for eight senescence-associated genes: IL-6, IL-8, MCP-1, Lamin B1, LAP2α, NQO1, PAI1, and p21. Outliers identified based on a z-score threshold $|z| > 2.0$, as defined in the data analysis section.

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